

13.1 TYPES OF DATA

Define data and data collection

We have learnt in previous class that the information collected from any field of study is called data. "A data is set of information and facts which is collected in the form of figures by observation and measurement.

We can collect data by several ways i.e observation, survey, interview, questionnaire etc.

We can analyze and represent the data to derive the result that helps us to examine the given information.

In routine surveys, we collect the information by using a questionnaire like

Tick (✓) your favourite food.

Biryani ☐

Tikka ☐

Fried Fish ☐

Karahi ☐

Few more examples of data:

- Information of favourite game
- Information of favourite subject of students of any school
- Information of most eaten food in any restaurant etc.

We represent the collected data or information in diagram or graphs and interpret it

In previous class we have learnt tally chart which is used for representing simple data.

Teacher's Note

Teacher should develop the questionnaire and complete it with the help of students in the classroom.

**Activity**

List the months of the year.

Ask each student in the class which months they were born.

Make a tally chart of this information.

Solution:

Month	Tally marks	No. of students
January		
February		
March		
April		
May		
June		
July		
August		
September		
October		
November		
December		

Teacher's Note

Teacher should complete the table by involving students.

Distinguish between grouped and ungrouped data

Data which shows the individual information is known ungrouped data.

For example:

Students marks in English test are given in the following table.

Name	Ali	Sana	Umer	Ahmed	Raza	Rahila	Saba	Maria	Faraz	Huma
Marks	46	50	30	42	69	59	70	75	40	72

If we arrange or classify the data in groups, it is called grouped data.

Example:

Following are the marks scored out of 100 by 15 students of class VI in test of Science.

76, 51, 65, 62, 70, 63, 69, 62, 75, 61, 80, 55, 65, 59, 60.

The student marks range from 50 to 80.

The marks can be organised into following groups.

51 – 60, 61 – 70, 71 – 80

Marks	Marks obtained	No. of Students
51 – 60	51, 55, 59, 60	4
61 – 70	61, 62, 62, 63, 65, 65, 69, 70	8
71 – 80	75, 76, 80	3

Represent the above data using tally chart.

EXERCISE 13.1

1. Define data.
 2. What is difference between grouped and ungrouped data?
 3. Which table is showing a grouped data?
- (i) Total marks secured by 12 students in an examination.

Names of Student	Ali	Sana	Sara	Adil	Amir	Asma	Huma	Arif	Babar	Anis	Anil	Umar
Marks	581	786	678	725	788	580	690	780	599	509	619	560

- (ii) Raza Book Store's sale in a month is shown by the table:

Number of books sold	Days
512 – 611	8
612 – 711	5
712 – 811	4
812 – 911	9

- (iii) Visitors in Sindh Museum.

Number of visitors	Days
150 – 199	3
200 – 249	4
250 – 299	1
300 – 349	2
350 – 399	3
400 – 499	2

- (iv) Runs scored by players of our school cricket team in one day match.

Player	1	2	3	4	5	6	7	8	9	10	11
Runs	62	41	15	59	22	10	8	2	43	7	21

13.2 BAR GRAPH

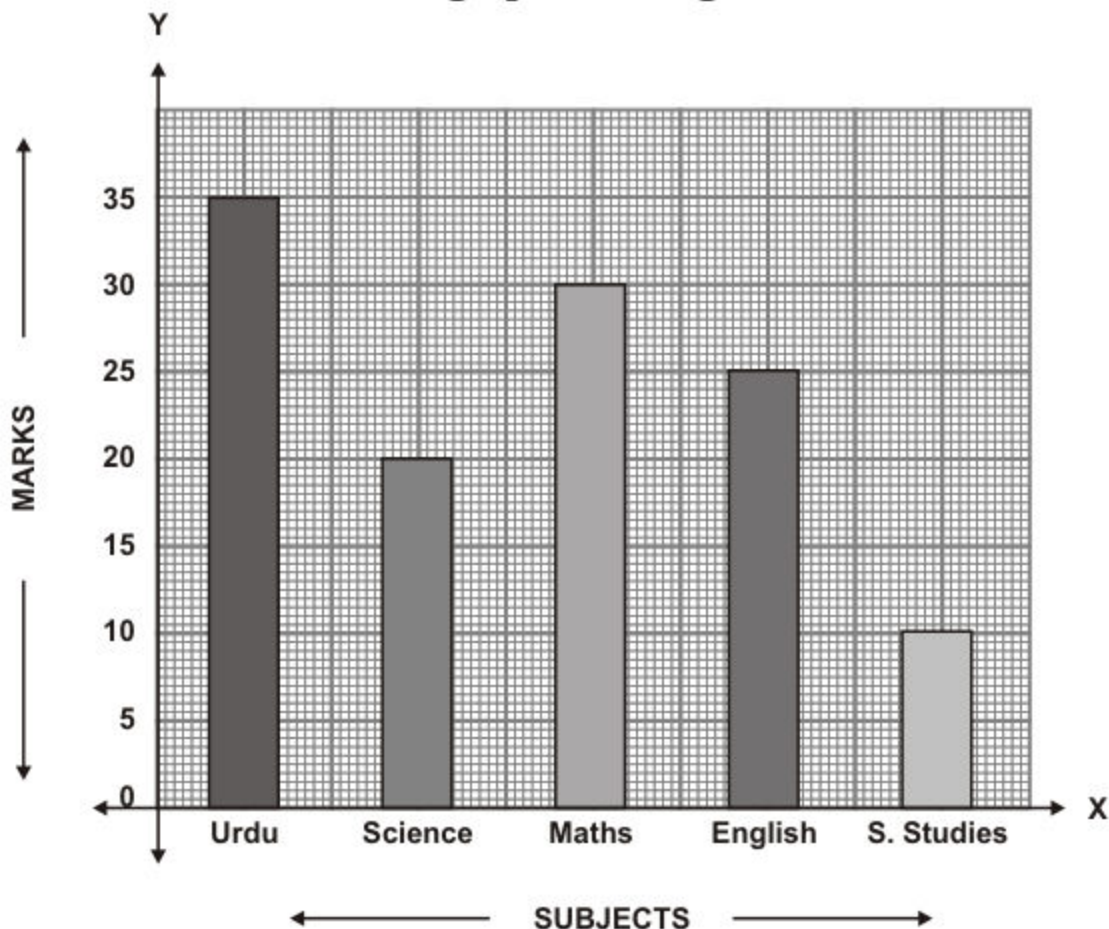
Draw horizontal and vertical bar graphs

We have learnt about Bar Graph in previous classes. A lot of information can be represented by a bar graph. In bar graph data is represented by a number of rectangular bars of the same width with equal spacing. We use two ways to draw bar graph.

- 1- Vertical Bar Graph
- 2- Horizontal Bar Graph

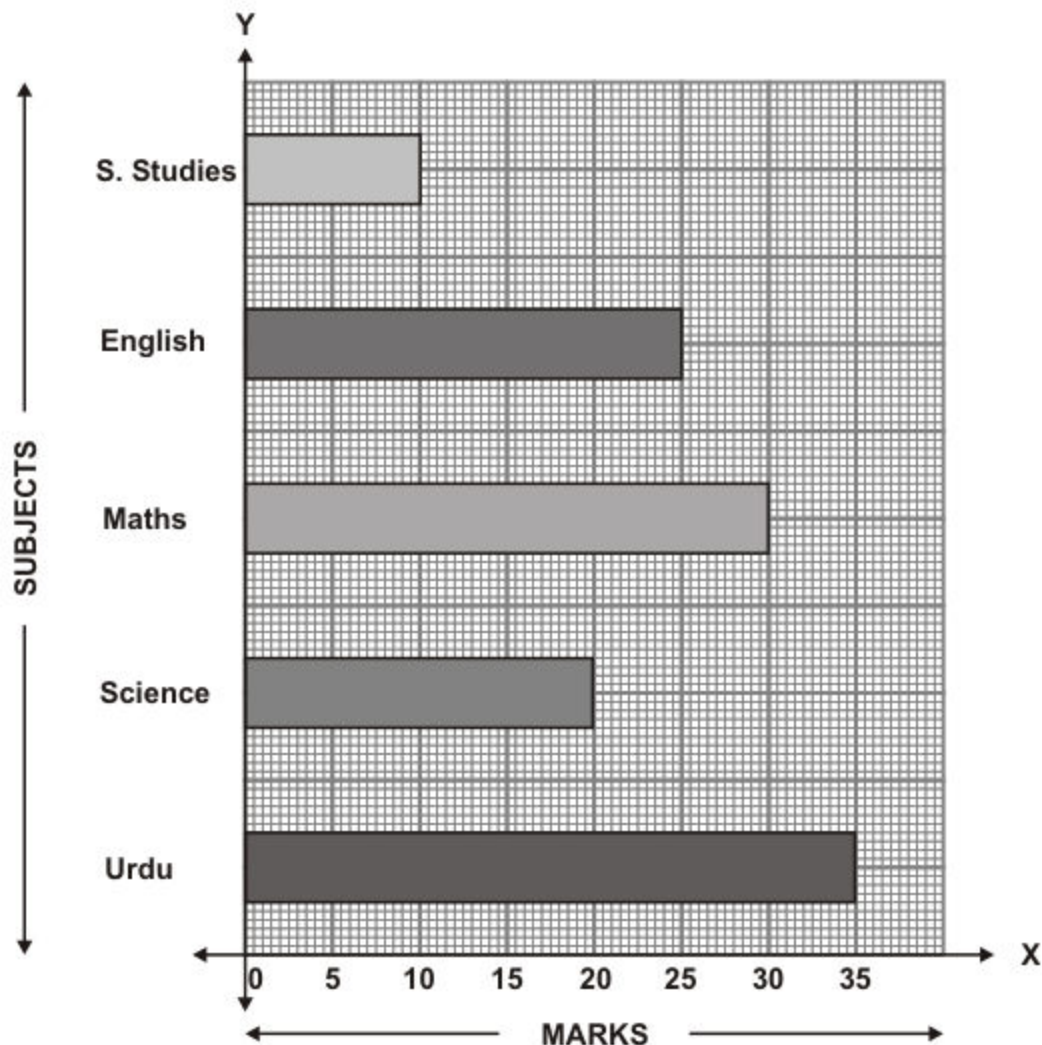
This bar graph represents the test marks of Fozia in different subjects through vertical bars.

Vertical bar graph showing marks of Fozia



This is the horizontal bar graph represents the same test marks of Fozia in different subjects.

Horizontal bar graph showing marks of Fozia



Steps to draw horizontal and vertical bar graphs.

Step 1. Take a graph paper and draw two lines $\overleftrightarrow{OX}$ and $\overleftrightarrow{OY}$, perpendicular to one another. $\overleftrightarrow{OX}$ is called **X-axis** and $\overleftrightarrow{OY}$ is called **Y-axis** and **O** is the **origin**. Give the graph a title.

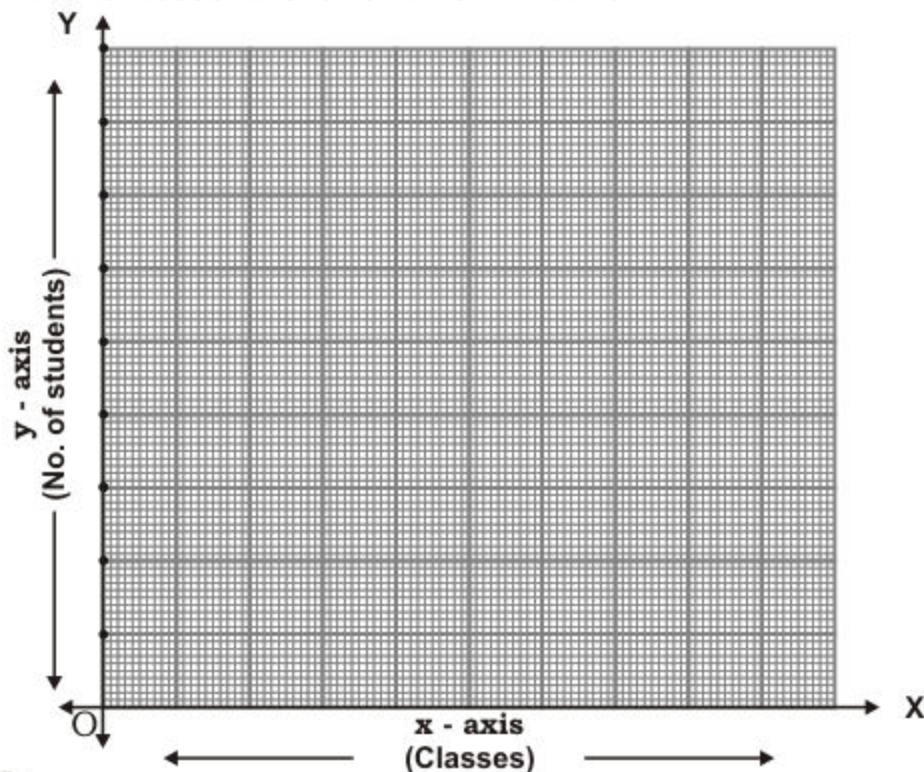
Step 2. Decide a scale for each axis.

Step 3. Make bars of equal breadth at equal distances according to the information.

Example: Draw a vertical bar graph showing the following information.

Class	I	II	III	IV	V
Number of students	45	30	25	36	45

Step I: In this graph the “number of students” are shown on Y-axis and classes are shown on X-axis.



Step II:

Scale: Two small square on Y-axis represents one student.

Length of each bar is taken according to the number of students.

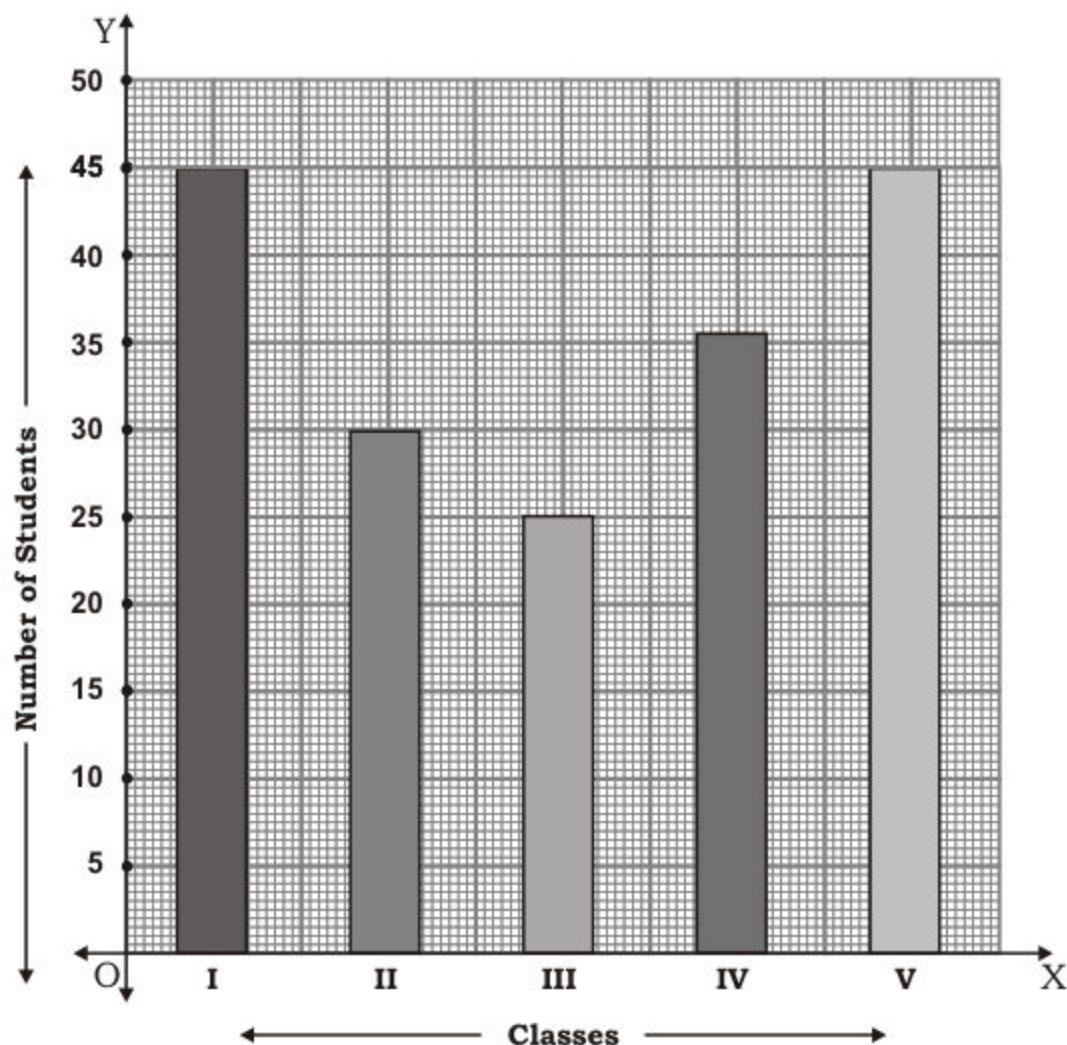
Breadth of each bar is 10 small squares and distance between them is 10 small squares on X-axis.

Step III:

In Class I: There are 45 students. Every two small squares on Y-axis represents a student, so we draw the bar graph up to 45 square to represent number of student.

Similarly, draw bars for representing other classes upto the required heights.

Vertical Bar Graph representing number of students in classes I to V.



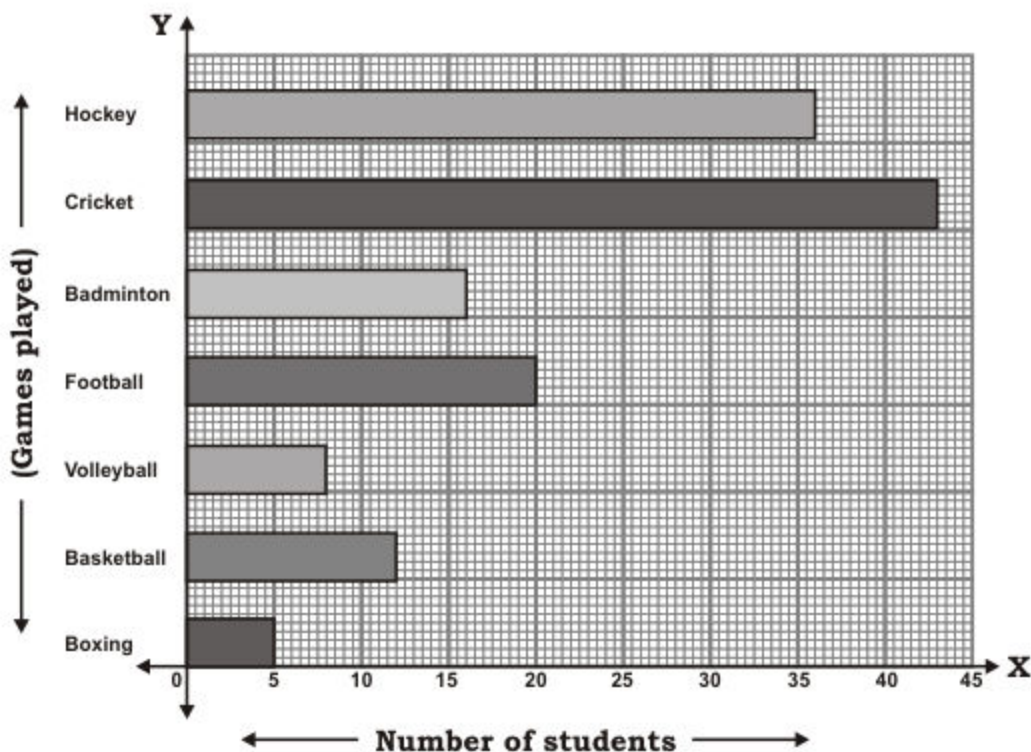
Check list for drawing the bar graph.

1. Read the information carefully.
2. Select proper axes for given information.
3. Make a scale.
4. Draw the bar for each information.
5. Label the graph and give a title.

Example: Games played by students of Class V in a school are given in the table below. Represent data in horizontal bar graph.

1. Read the table, there are seven different types of games played by students.
2. We select x-axis for number of students and y-axis for games played.
3. Make a scale for number of students and played games. 1 unit represents 2 students.
4. Draw horizontal bar for each game.
5. Label both axis and give the graph a title.

Game played	No. of students
Hockey	36
Cricket	43
Badminton	16
Football	20
Volleyball	8
Basketball	12
Boxing	5



**Activity**

Write three sentences about the information you get from the previous graph.

1. Cricket is played by most of the students than any other game. So the longest bar will represent cricket with length of bar at 43 units.
2. _____
3. _____

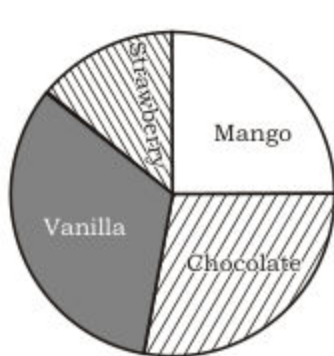
13.3 PIE GRAPH

A pie chart is circle graph. When we have many quantities in data, we used a pie graph or pie chart to represent it.

Read a pie graph

In this graph circle is used to represent data. Different quantities are represented by sectors. We know that there are 360° in a circle. The quantities are represented by sectors of a circle. It is useful to show or to compare various quantities at a glance.

Example 1: The data of cone ice cream of different flavours sold at a shop in a day are shown in the pie graph as under.



Represents mango flavour.



Represents chocolate flavour.



Represents vanilla flavour.



Represents strawberry flavour.

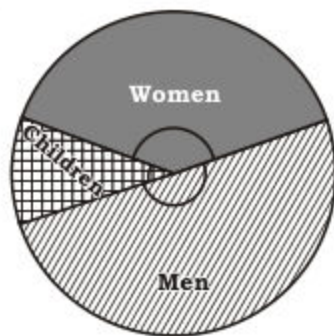
Study the graph and answer the following questions:

- (i) Name the flavour with most sale. Vanilla
- How do you come to know? Because the representing sector is largest in size.
- (ii) Name the flavour with least sale. Strawberry
- How do you come to know? Because the representing sector is smallest in size.
- (iii) Select the flavour which is sold more from mango and vanilla. Chocolate



Activity

The number of 600 pilgrims in a plane is shown by the following Pie graph.



Here



represents men



represents women



represents children

Read the pie graph and answer the following questions:

- Who are the most on the plane? Men
- Who are least on the plane?
- Are there children equal to women passengers?
- Are men passengers more than women passengers?

EXERCISE 13.2

1. Draw a vertical bar graph to represent the information given below:

Monthly Sale of Fans in a Shop

Name of Months	January	February	March	April	May	June
No. of Fans	70	50	35	60	90	95

2. Represent the following information in a vertical bar graph.

Monthly Expenses of a Akbar's Family

Items	Food	Education	Rent	Telephone	Electricity	Gas
Expenses (in rupees)	6000	5000	4500	1500	2000	300

3. Represent the following information in a horizontal bar graph.

Sale of Variety Show Ticket

Name of cities	Hyderabad	Sukkur	Larkana	Dadu
Number of Tickets sold	10,000	6,000	3,000	9,000

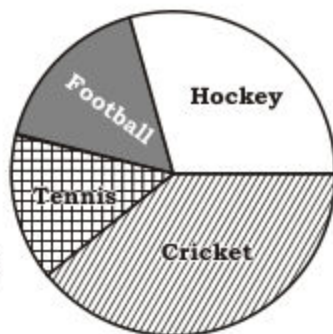
4. Draw a horizontal bar graph showing the marks obtained by Amjad during annual examination.

Subject	Islamiat	English	Urdu	Maths	S.Studies	Science	Sindhi
Marks Obtained	70	60	65	90	45	55	80

5. Following pie graph is representing the games played by the students of class VI.

Study the pie graph and answer the following questions.

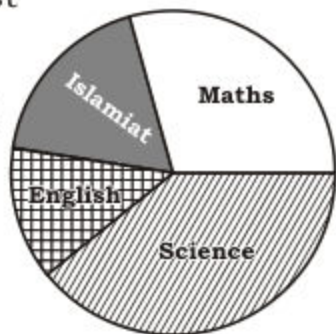
- Which game is played less?
- Which game is played most?
- Which games are played equally?



6. The marks obtained in different subjects by Aftab are shown by the following pie graph.

Study the pie graph and answer the following questions.

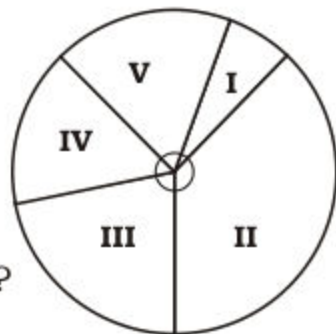
- In which subject, Aftab secured highest marks?
- In which subject, Aftab secured lowest marks?
- In which subject, Aftab secured more marks in science or maths?



7. The following pie graph shows the attendance of the students of different classes of a school on Saturday.

Study the pie graph and answer the following questions.

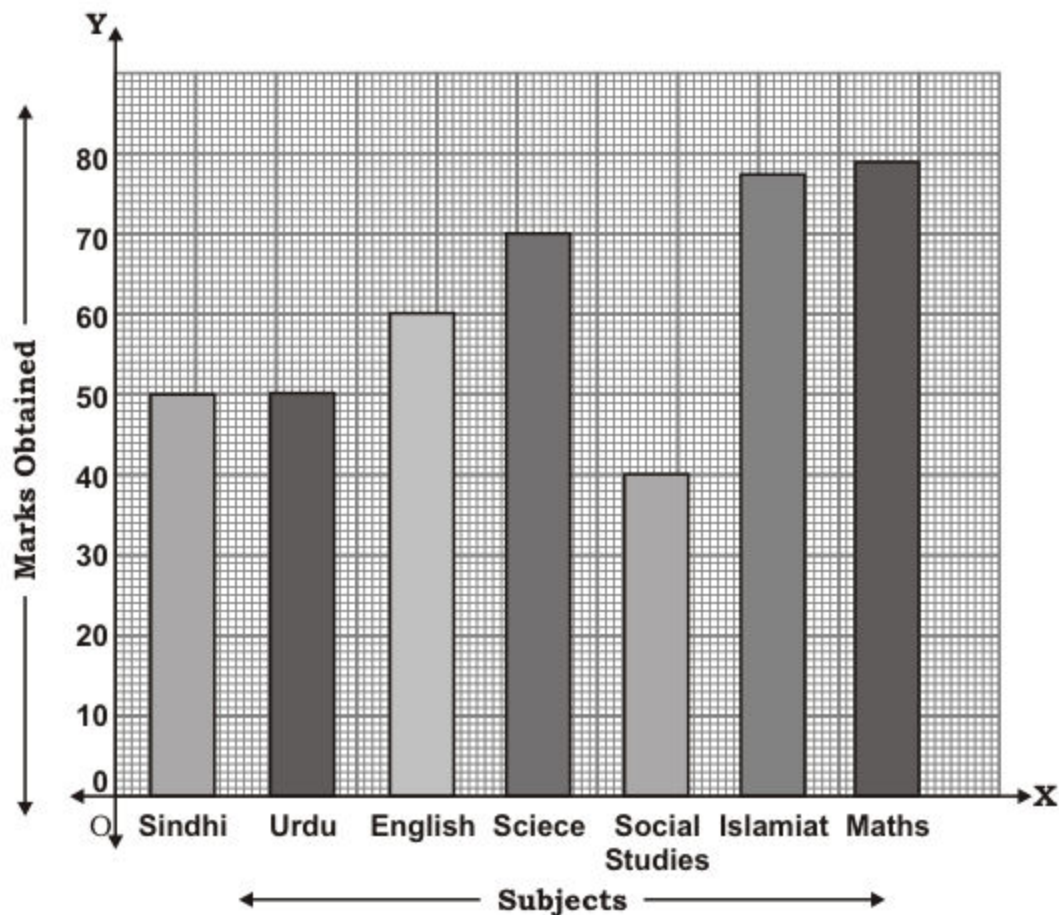
- In which class, attendance was the highest?
- In which class, attendance was the lowest?
- Which classes have same number of students?



REVIEW EXERCISE 13

1. The marks obtained by Saira are shown by a vertical bar graph. Answer the given questions by reading the bar graph.

Graph represents marks obtained by Saira in different subject.



- In which subject Saira obtained highest marks and how many?
- In which subject Saira obtained lowest marks and how many?
- In which subjects Saira obtained equal marks and how many?
- Prepare a chart which represents the marks in each subject.

2. Are the following statements true or false?

- (i) The reading of numerical data in pictograph is easy.
- (ii) In bar graphs numerical data is represented by the rectangular boxes of uniform width.
- (iii) To arrange the data in some order is called ungrouped data.
- (iv) A rectangular diagram is called pie chart.

3. Fill in the blanks.

- (i) To represent numerical data pictorially is called _____ graph.
- (ii) Representation through _____ graphs is easier and better than the pictograph.
- (iii) In bar graphs, bars can be drawn either horizontally or _____.
- (iv) _____ is defined as a set of facts.
- (v) Numerical facts which are obtained from field is known as _____.

SUMMARY

- Data is defined as “a set of observation, measurements or facts”.
- Data is called Group Data if classified under certain criteria otherwise it is ungrouped Data.
- In bar graph the quantities are represented by bars of equal width.
- In a bar graph, the bars may be vertical or horizontal.
- In a Pie graph, a circle is divided into a number of sectors, with one sector for each category of Data.